

Module B

Krauss, Do, Huck (2017)

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Papers and materials

Krauss, C., Do, X. A., & Huck, N. (2017).

Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500.

European Journal of Operational Research, 259(2), 689–702.

Live: `notebooks/02_krauss_simplified.ipynb`

Paper card: Krauss, Do & Huck (2017)

	<i>European Journal of Operational Research 259: 689–702</i>
Question	Can deep nets, gradient-boosted trees and random forests rank S&P 500 stocks well enough for profitable statistical arbitrage?
Data	S&P 500 constituents from month-end membership lists (no survivor bias); daily data 1990–2015; trading December 1992 – October 2015.
Method	Classify whether tomorrow's return beats the cross-sectional median. Features: 31 lagged returns (1–20 days, then monthly out to 240 days). DNN, GBT, RF and their equal-weighted ensemble.
How it is tested	Walk-forward: train 750 days, trade the next 250, 23 periods. Each day, go long the top 10 and short the bottom 10.
Main result	The ensemble earns 0.45% a day before costs and 0.25% after 5 bp per half-turn. RF beats the DNN. Returns decay after 2001.
Watch for	The abstract claims a challenge to semi-strong efficiency. Which form of efficiency do lagged returns test?
Code & data	Datastream data (licensed); no code package. Our notebook rebuilds the design on public Yahoo data.

Contrast with Gu

Gu–Kelly–Xiu	Krauss–Do–Huck
Predict the return (regression)	Predict above the CS median (classification)
Characteristic zoo, monthly	31 lagged returns, daily
OOS R^2 + sorts	Probabilities → ranked long–short

Same ML tools, different **decision object**.

Economic question

Can classifiers rank S&P 500 stocks by their chance of **beating the cross-sectional median tomorrow**, well enough to build a market-neutral long–short book?

Hypothesis check (5 min)

The abstract: "our findings pose a severe challenge to the **semi-strong** form of market efficiency."

The features: **31 lagged returns**, nothing else.

Which form of efficiency does this test?

Restate the paper's hypothesis in one sentence.

Label & sample (paper)

- S&P 500, daily data 1990–2015, trading **Dec 1992 – Oct 2015**
- Survivor bias removed with **month-end constituent lists** (§3.1)
- Label (§4.2): $y_{i,t+1} = 1\{r_{i,t+1} > \text{median}_j r_{j,t+1}\}$, **not** measured against the index
- The median label gives balanced classes and a market-neutral target

Features & validation

- $R_{t,m}$ for $m \in \{1, \dots, 20\} \cup \{40, 60, \dots, 240\}$: daily for one month, monthly out to a year
- Study period: **750 days training + 250 days trading**, 23 non-overlapping trading periods
- **No hyperparameter search**: a heuristic DNN architecture, H2O defaults elsewhere
- Rank by $P(y = 1)$, not by the hard 0/1 label

Model ladder (paper)

Model	Key settings	Daily return, $k = 10$, pre-cost
DNN	31-31-10-5-2, maxout, dropout	0.33%
GBT	100 trees, depth 3, lr 0.1	0.37%
RF	1,000 trees, depth 20	0.43%
ENS1	equal-weighted average	0.45%

Classroom: Logistic → RF → GBT → small MLP.

From scores to P&L

Each day:

1. Predict probabilities
2. Long top- k , short bottom- k ($k = 10$ of about 500)
3. Track the excess long–short return
4. Costs: **5 bp per half-turn** cut ENS1 from 0.45% to **0.25% per day**

A great AUC with fatal turnover is not a strategy.

Decay

- Returns spike in turmoil (the dot-com bust, 2008)
- They **decline after 2001**, as ML and computing spread

Link: McLean & Pontiff (2016, *JF*): publication erodes predictability.

Is the edge a market fact or a period fact?

Our real mode is not a replication

	Paper	Our real mode
Universe	~500, historical members	50 current names (survivors)
k per side	10 (2%)	set $k = 1$ to match 2%
Membership	month-end lists	none

In our run: AUC ≈ 0.50 and the best break-even cost is ≈ 3 bp, below the 5 bp cost.
Survivorship lifts the *level* of returns, not the edge in *ranking* them.

Live notebook agenda

1. Build labels (**above the cross-sectional median**)
2. Fit Logit / RF / GBT / MLP
3. Long–short from ranks, k scaled to the universe
4. Cost grid in bp; break-even cost
5. Experiment: change k ; median vs index label

Discussion hooks

- Can AUC rise while Sharpe falls?
- What does high turnover say about capacity?
- After costs: would you still ship the NN?
- Nothing was tuned. Does that make the horse race fairer or less fair?

Frontier pointer

Guijarro-Ordóñez, J., Pelger, M., & Zanolini, G. *Deep Learning Statistical Arbitrage. Management Science*, 72(9).

The same question, with economic structure:

- trade **residual** portfolios after removing factor exposure
- extract the signal with a CNN + transformer
- learn the **trading policy** directly, with costs inside the objective

The same step as Gu → Chen–Pelger–Zhu: put the objective inside the loss.

Takeaway

A classifier is only as good as the **label, the portfolio rule and the frictions** you attach to its scores.

Next: Module C, where credit decisions and the benchmark protocol are the contribution